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AI Agents for Liquid-Cooled AI Factories

Improving Thermal Stability via Proactive AI Control

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Table of Contents

Executive Summary **3**
Key Findings: 3
Challenge: Thermal Inertia and Control Latency **4**
The Transient Nature of the AI Factory 4
Feedback Control and Transport Delay 4
Safety Buffer by Sub-Cooling 4
Methodology: AI-Driven Feed-Forward Control **6**
Mechanism of Action: Power as a Leading Indicator 6
Supervisory Safety 6
Validation: Performance with NVIDIA and CoreWeave **7**
NVIDIA Testing Results 7
CoreWeave Testing Results 8
Self-Learning Capability 11
Future: Pathway to Max-Q AI Factories **12**
Acknowledgements **14**

Executive Summary

Modern AI data centers (i.e. AI factories) have entered a new era of scale and density — defined not just by higher power requirements, but by the volatility of the workloads themselves. As facilities scale to accommodate tens of thousands of networked GPUs, they face a new operational challenge: managing the thermal shock of rapid and coordinated power ramps.

Unlike traditional CPU workloads, AI training workloads are highly-synchronized. When a large scale training or inference job starts, the IT load can ramp up significantly within a short time period. This creates rapid heat generation that traditional control techniques, specifically Proportional-Integral-Derivative or PID controls, are too slow to catch.

To mitigate the risk of GPU thermal throttling (which reduces token throughput) — and maximize SLAs for end users — facility operators currently run liquid cooling systems at artificially low temperatures. This sub-cooling strategy protects GPU performance by providing a thermal buffer to absorb temperature spikes. However, sub-cooling arrives at the cost of operational efficiency (e.g. power usage effectiveness, or PUE) and limits the facility's ability to maximize compute capacity.

In a world where computational resources are limited by energy availability — and grid interconnection times take many years — every watt that isn't being used for valuable token generation is a wasted watt.

This whitepaper by Phaidra, NVIDIA, CoreWeave, and Applied Digital presents a validated methodology, on production NVIDIA Grace Blackwell platforms, for agentic AI control of liquid-cooling Coolant Distribution Units (CDU). By shifting from reactive, temperature-based feedback loops to proactive, feed-forward control, Phaidra's AI Agents can predict and largely eliminate thermal spikes before they occur.

This approach maximizes token production (by reducing the likelihood of GPU thermal throttling) while unlocking additional power for revenue-generating IT compute.

Key Findings:

- **Validation:** Joint testing with NVIDIA and CoreWeave, on production NVIDIA Grace Blackwell clusters with live AI workloads, confirmed a ~75% reduction in thermal spike overshoot compared to an optimally-tuned PID baseline. These results were consistent across both in-rack and end-of-row CDU architectures.
- **Mechanism:** By utilizing rack power data to predict thermal response, the AI Agent reduced the control delay from minutes to <10 seconds.
- **Impact:** This stability allows facilities to (1) maximize token production by minimizing thermal throttling, and (2) safely raise supply water temperatures, which significantly reduces the cooling load and unlocks stranded capacity for revenue-generating IT compute.

Challenge: Thermal Inertia and Control Latency

To understand why traditional cooling controls fall short in AI Factories, we must first understand how the workload differs from traditional data centers.

The Transient Nature of the AI Factory

Data center thermal management has historically been designed for steady-state or slowly shifting loads, like web traffic. However, large data centers with AI workloads — commonly known as AI Factories — have fundamentally different load profiles. Modern GPU clusters function as a synchronized single unit of compute. When a training job is dispatched, thousands of GPUs (to hundreds of thousands in the largest clusters) ramp up simultaneously. The resulting power profile is "peaky," characterized by 20-100% load ramps (idle-to-max capacity) within seconds.

Applied Digital operates large-scale AI Factories where these rapid power ramps create complex interactions between power distribution, liquid cooling systems, and GPU workloads across thousands of racks simultaneously. Maintaining thermal stability while maximizing the amount of available power that can be directed towards compute becomes a critical operational challenge.

Because heat generation is a direct function of power consumption, these power spikes rapidly translate into thermal spikes. If the cooling system cannot reject this heat fast enough, the coolant temperature rises rapidly, breaching the GPU's thermal limit (T-Limit) and forcing the hardware to throttle performance to protect itself.

The industry currently relies on traditional PID control loops to actuate CDU valves to maintain the supply temperature at a desired setpoint. PID is a robust and time-tested method of controlling hardware to a desired setpoint in slow-moving systems. However, the load changes in AI factories are too large and too sudden for traditional PID controllers to successfully maintain the temperature setpoint during these power spikes.

Feedback Control and Transport Delay

A PID controller relies on sensor feedback. For liquid-cooling CDUs, the PID controller waits for a sensor to register a change in coolant supply temperature before it modulates the energy valve. The magnitude of the actuation is proportional to the error signal (i.e., the discrepancy between the measured versus target temperature setpoint).

- **The Physics of Delay:** Coolant has a high thermal inertia. By the time the fluid absorbs heat from the GPU and registers a sufficiently large temperature deviation to trigger the PID loop, the thermal spike has already begun and will continue climbing. There is typically a 3-5 minute delay between when a load ramp occurs and the subsequent thermal spike.
- **The Consequence:** Even with optimal tuning, traditional PID loops on high-density racks result in significant thermal overshoot during rapid load ramps.

Safety Buffer by Sub-Cooling

Because operators know their existing control systems cannot react fast enough to catch these temperature spikes, they compensate by lowering the baseline temperature. For example, although

NVIDIA's Technology Cooling System (TCS) water specifications allow for higher operating temperatures (up to 45°C), many facilities running NVIDIA AI infrastructure operate significantly lower, around 20-30°C, to create a safety buffer.

This approach is very expensive:

- **Energy Waste:** Sub-cooling places a massive burden on the facility-level chiller plant. Research from the Open Compute Project (OCP) indicates that raising coolant temperatures from 30°C to 40°C can reduce annual chiller energy consumption by roughly 50% across major ASHRAE climate zones. See **Figure 1** below.
- **Operational Drag:** Maintaining optimally-tuned PID loops to maintain these best-case dynamics is a labour-intensive and unscalable task as facility complexity grows.

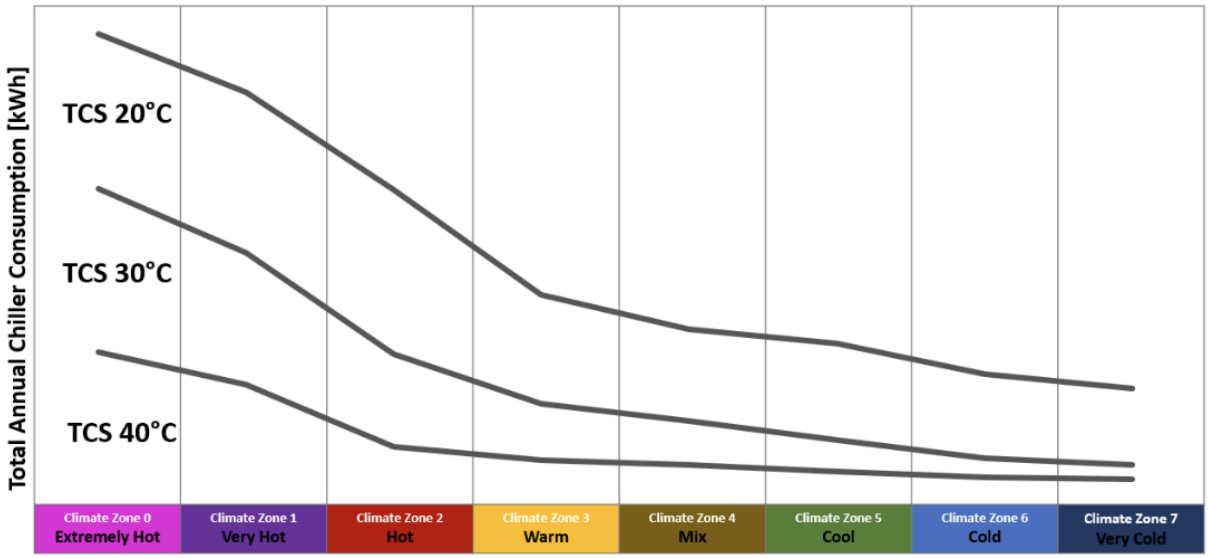

Figure 1 - [Open Compute Project: "30C coolant - A durable roadmap for the future (published Oct 2024)."]

While the energy cost of over-cooling AI factories is high, the opportunity cost (i.e. revenue impact) of this excess power consumption is even higher. Table 1 below shows the maximum cooling and IT capacities of a baseline 1 gigawatt (GW) AI-factory with 1.43 design PUE (assuming 25MW of non-cooling overhead), alongside the incremental revenue unlocked by freeing up cooling power for revenue-generating IT power.

TCS temperature (C)	Max Cooling (MW)	Max IT (MW)	Additional IT Capacity (MW)	Additional Revenue ($bn/yr)
25 (design)	275 (design)	700 (design)	-	-
30	236.5	738.5	38.5 (+5.5%)	$2.2 billion
35	207.6	767.4	67.4 (+9.6% gain)	$3.8 billion

Table 1 - Additional revenue impact on a baseline 1GW AI factory using modular air-cooled chillers (1.43 design PUE).

The calculations above assume a conservative 20% cooling energy reduction from TCS 25→30C, and a 35% reduction from TCS 25→35C. We further de-rate the additional IT capacity by 30% since it must be cooled. Finally, we use the standard metric of a 1GW AI factory generating $200bn in revenue over 5 years (i.e. $40bn per year) to estimate the additional revenue impact. These figures are for illustrative purposes only and do not constitute financial forecasts or guaranteed returns.

In summary: the immediate impact of eliminating thermal spikes is minimizing GPU throttling, improving SLAs for end users, and reducing the human labor required for CDU thermal management.

The broader impact is increasing AI factory token throughput. This is achieved by reallocating the cooling energy savings from safely increasing TCS temperatures for additional IT capacity — thereby generating significantly more revenue from higher token throughput.

Methodology: AI-Driven Feed-Forward Control

To solve the latency gap, Phaidra has developed a reinforcement learning (RL) AI Agent that sits at the edge and controls the CDU secondary supply setpoint directly.

Mechanism of Action: Power as a Leading Indicator

The core innovation is shifting from reactive feedback control (based on temperature changes), to proactive feed-forward control based on power telemetry. The key stages are outlined below:

1. **Ingestion:** The Agent ingests real-time Rack Power Data and CDU parameters at low latency time steps.
2. **Inference:** Utilizing a proprietary RL algorithm, the Agent identifies IT load spikes the moment they occur on the power bus and determines the optimal action (via a learned policy) to prevent the expected future thermal spike.
3. **Action:** The Agent generates a control signal to the CDU to dynamically adjust the supply setpoint before the heat spike registers in the fluid temperature. This enables the onboard CDU controller to modulate the energy valve proactively.

By using power as the trigger, the system reduces the effective response delay from 3-5 minutes with traditional PID control on the supply temperature, to under 10 seconds.

Supervisory Safety

Crucially, the AI Agent acts as a supervisory control system. It does not replace the underlying hardware safety mechanisms nor existing PID control system. Instead, the AI generates optimal, feed-forward setpoint commands which are passed to the existing controller for execution, i.e. it acts as an intelligent supervisory layer on top of the existing controls.

- **Guardrails:** The Agent is bound by hard-coded allowable ranges. It cannot issue commands outside these safety limits.
- **Failover:** In the event of network loss or anomaly, the system automatically fails over to the standard local control mode, ensuring hardware safety is never compromised.

Validation: Performance with NVIDIA and CoreWeave

This methodology was validated with A/B testing in two separate production environments:

1. Live NVIDIA environment consisting of an NVIDIA DGX SuperPOD (with 8 DGX GB200 systems and 2 end-of-row CDUs) cluster running Megatron (LLM training) workloads.
2. Live CoreWeave environment consisting of (2) GB200 NVL72s with in-rack CDUs.

Following these validation results, Phaidra and CoreWeave are scaling the integration of AI-driven thermal management across CoreWeave's liquid-cooled infrastructure, focusing on the optimization of its NVIDIA Grace Blackwell (GB200 and GB300) deployments. The results from this large-scale, production rollout are consistent with the results shown in this paper and further reinforce CoreWeave's leading market position as the sole Platinum-certified GPU cloud (per SemiAnalysis ratings) with industry-leading reliability and end-user SLAs.

NVIDIA Testing Results

The system was tested against a baseline Vertiv XDU1350 (end-of-row) controller using PID control that had been rigorously tuned by NVIDIA to achieve optimal performance. It is important to note that the Vertiv XDU1350 CDU Controller remains the physical actuation mechanism. The Phaidra AI Agent calculates the optimal setpoint value for the controller to achieve.

- **Thermal Stability:** The AI Agent reduced the magnitude of thermal spike overshoot by ~75% for load ramps of 25–70% (equivalent to 250–700kW on a DGX SuperPOD).
- **Response:** As shown in the data below, the Agent pre-emptively adjusts the CDU temperature setpoint (yellow) in response to the instantaneous power spike (red), keeping the supply water temperature (blue) to the GPUs stable.

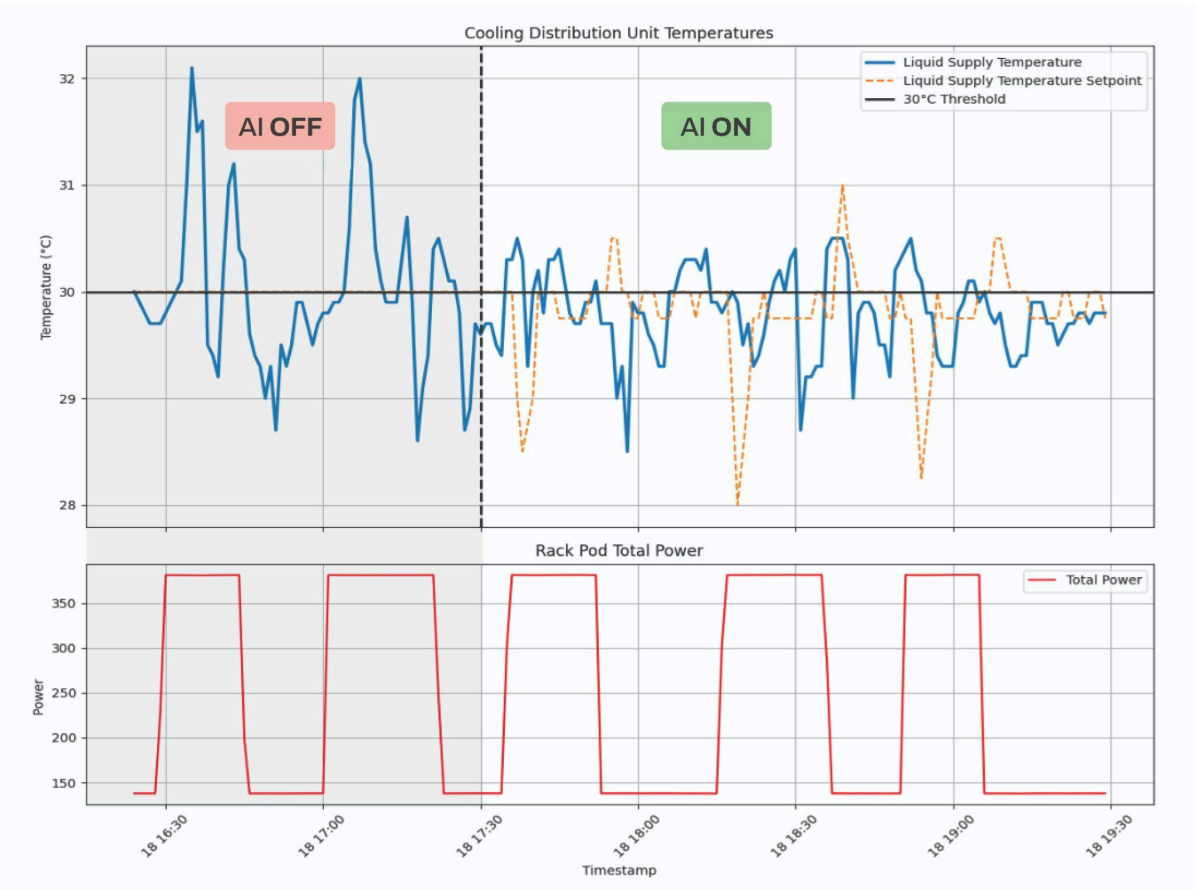

Figure 2 - This figure illustrates the reduction in thermal spike overshoot (blue) when the AI is On and recommends setpoints (yellow) to counteract the rack power spikes (red).

CoreWeave Testing Results

This test consisted of (2) NVIDIA GB200 NVL72 racks using Vertiv in-rack CDU controllers that had been rigorously tuned to achieve optimal performance. As before, the Vertiv CDU controller remains the physical actuation mechanism and receives optimal setpoint commands from the Phaidra AI agent. We show two power scenarios: 60kW load ramps and 100kW load ramps.

- **Thermal Stability (60kW spikes):** The AI Agent reduced the magnitude of thermal spike overshoot by ~75% for load ramps of 35 → 60kW (3-4C overshoot to 0.5-1C overshoot).
- **Thermal Stability (100kW spikes):** The AI Agent reduced the magnitude of thermal spike overshoot by ~80% for load ramps of 35 → 100kW (5-6C overshoot to 1C overshoot).
- **Response:** As before, the Agent pre-emptively adjusts the CDU temperature setpoint in response to the instantaneous power spike, keeping the supply water temperature to the GPUs stable.

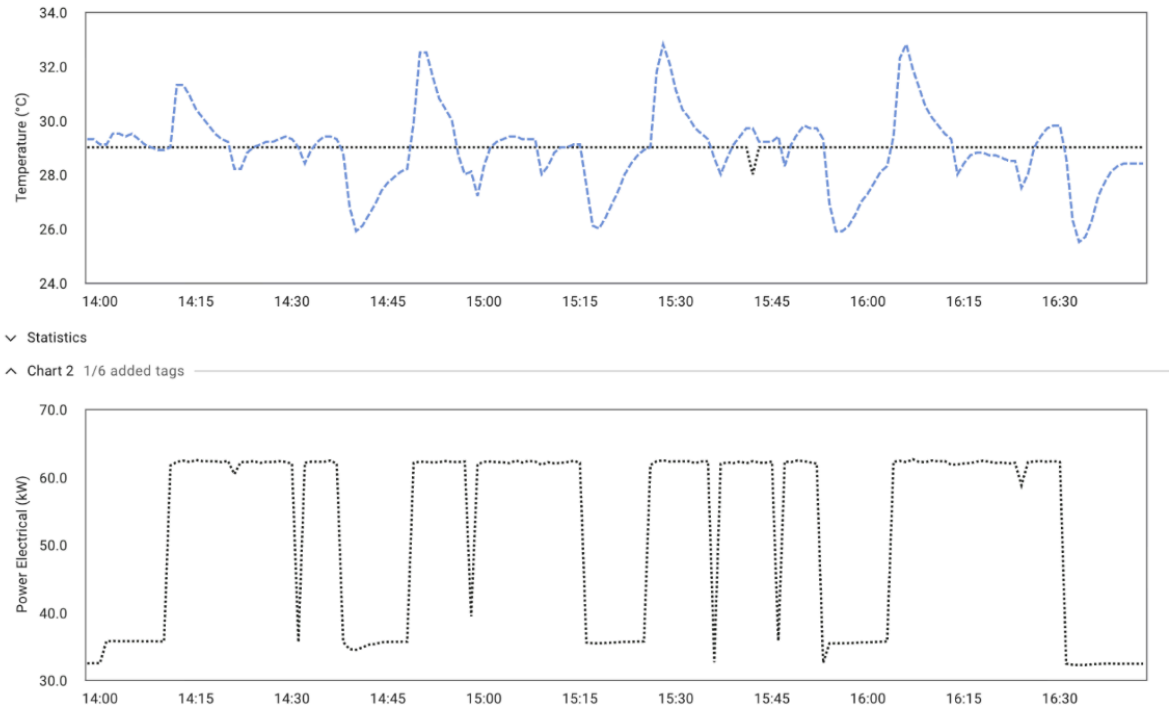

Figure 3 - Baseline (AI_OFF) behavior for 35 → 60kW load ramps, resulting in 3-4C thermal overshoot.

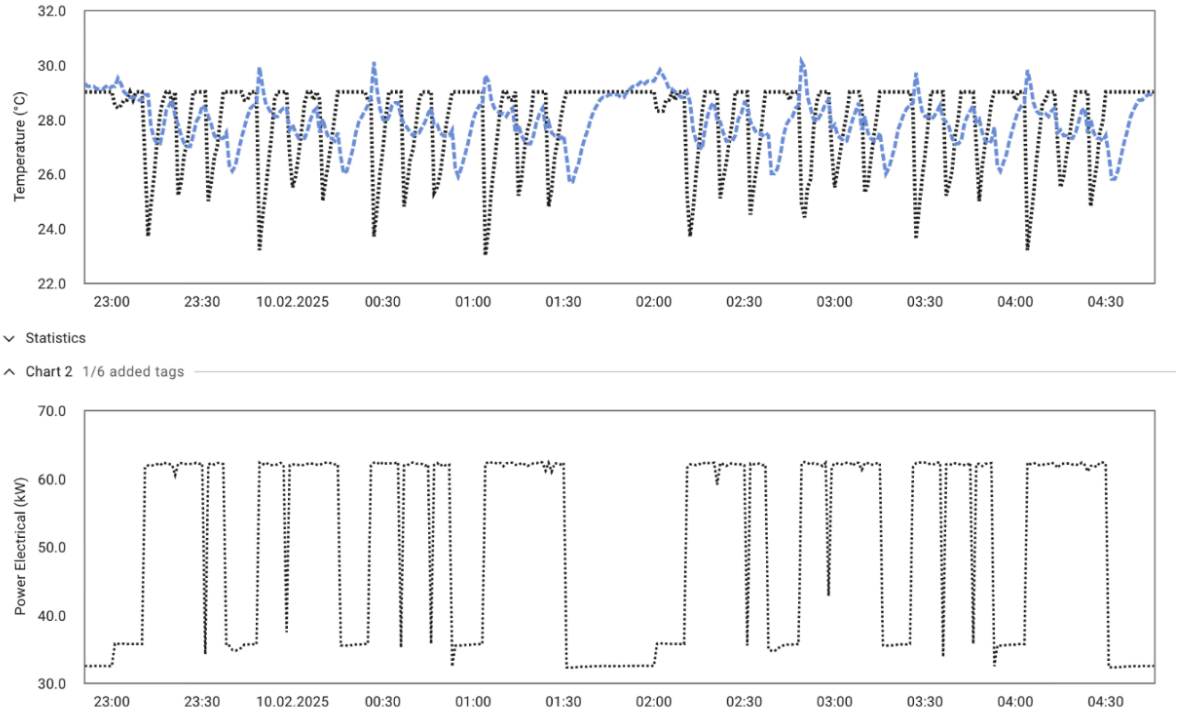

Figure 4 - AI_ON behavior for 35 → 60kW load ramps, resulting in 0.5-1C thermal overshoot.

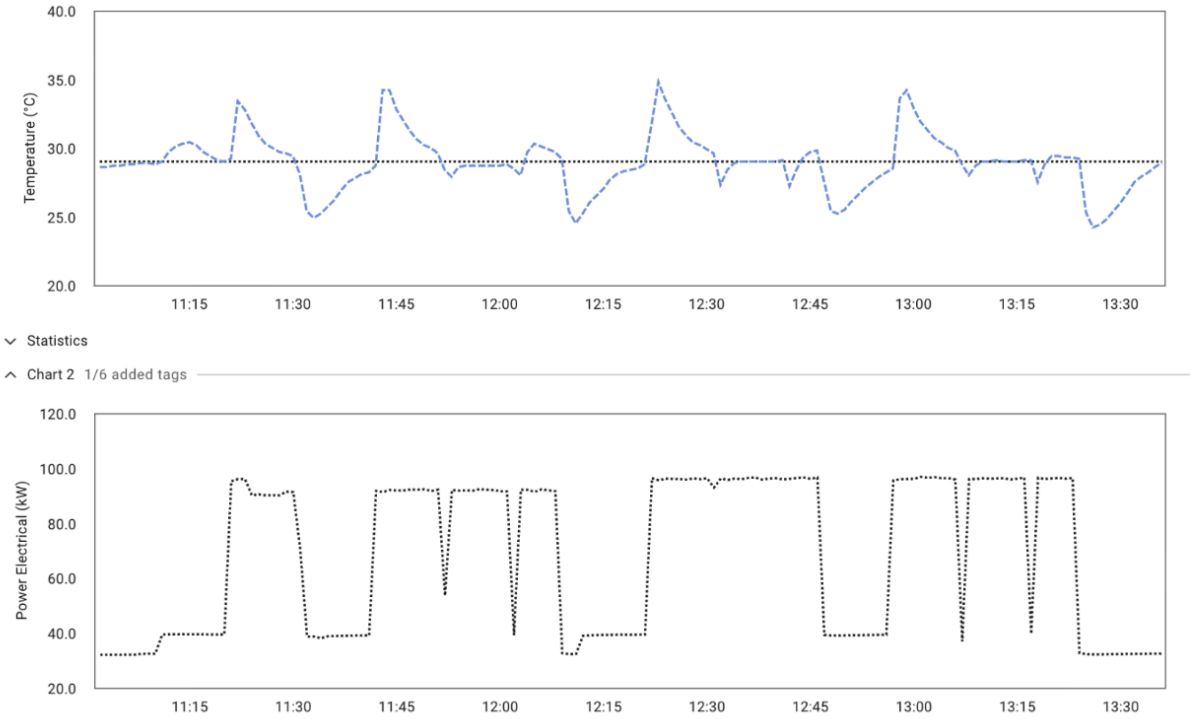

Figure 5 - Baseline (AI_OFF) behavior for 35 → 100kW load ramps, resulting in 5-6C thermal overshoot.

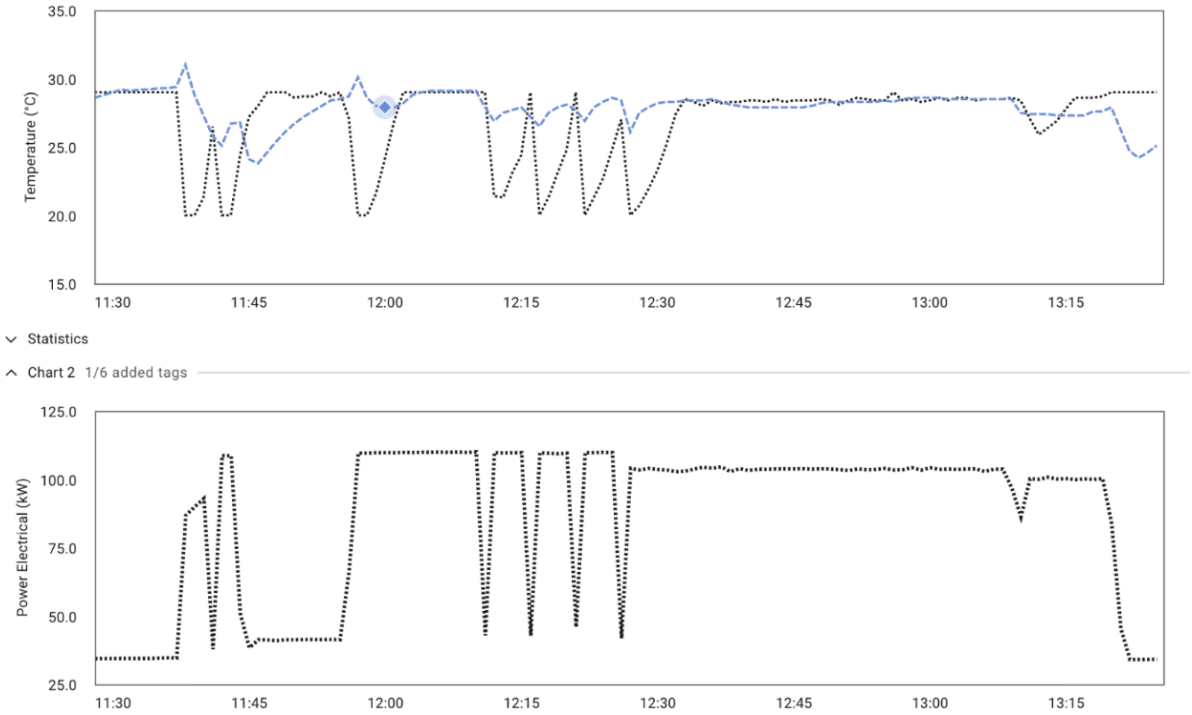

Figure 6 - AI_ON behavior for 35 → 100kW load ramps, resulting in 1C thermal overshoot.

The results shown here correspond to the NVIDIA and CoreWeave deployments outlined above. Future results may vary depending upon facility configurations and environmental conditions.

Self-Learning Capability

A major challenge in data center operations is that every facility is unique. This is compounded by the fact that CDUs from different manufacturers will respond to system changes with different response characteristics. Finally, the liquid cooling systems themselves will not always behave as they did when first commissioned — with wear and tear (and changing deployment patterns) over time, the same CDU will begin to respond differently.

Phaidra addresses these operational challenges via self-learning. Concretely, the AI agents teach themselves the optimal manner of controlling that CDU to achieve the desired performance. This enables Phaidra to scale DC capacity without the need for custom tuning (and re-tuning) of each CDU — saving time and scarce labor resources.

- **Pre-Training:** Phaidra pre-trains generalized Agents on an operational digital twin (i.e. CDU simulator) specific to each CDU manufacturer and secondary loop architecture. This enables the Agents to provide significant performance improvements over traditional PID control even with these initial, pre-trained parameters.
- **Post-training:** Upon live deployment in the new environment, each generalized Agent begins post-training on the unique site data and converges to optimal performance within several hours of deployment.

This technique of pre-training AI agents in simulation — and post-training on live operational data — enables Phaidra to scale production capacity quickly with key customers and partners like CoreWeave. The scaling technique is further detailed in this blog post outlining Phaidra's collaboration with NVIDIA to co-design AI agents for gigawatt (GW) scale AI factories for maximum token throughput (NVIDIA DSX Max-Q).

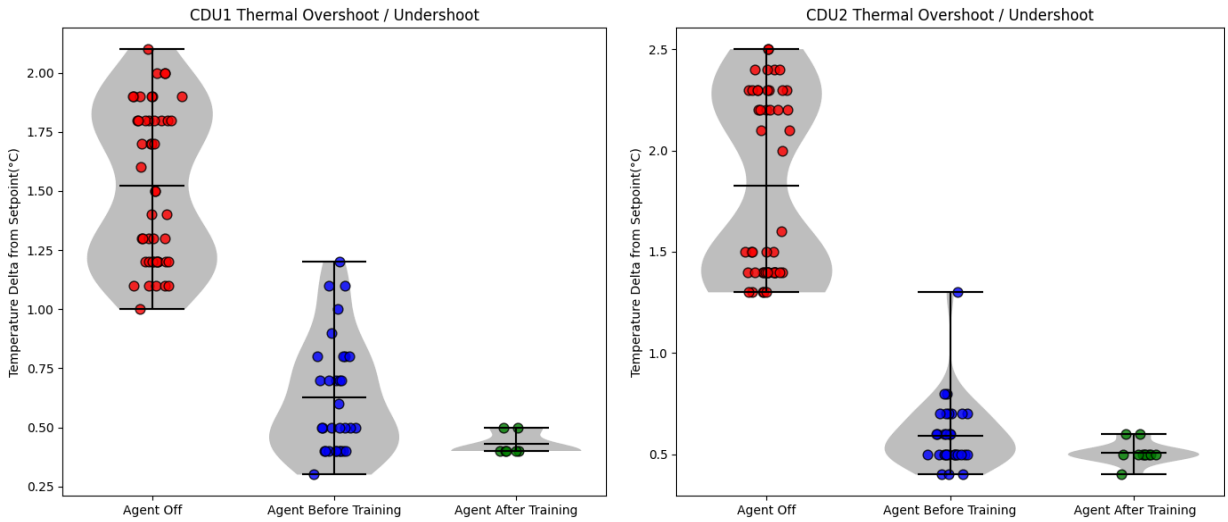

Figure 7 - A statistical analysis of AI agent learning across 3 scenarios (Agent Off, Pre-Training and Post-Training).

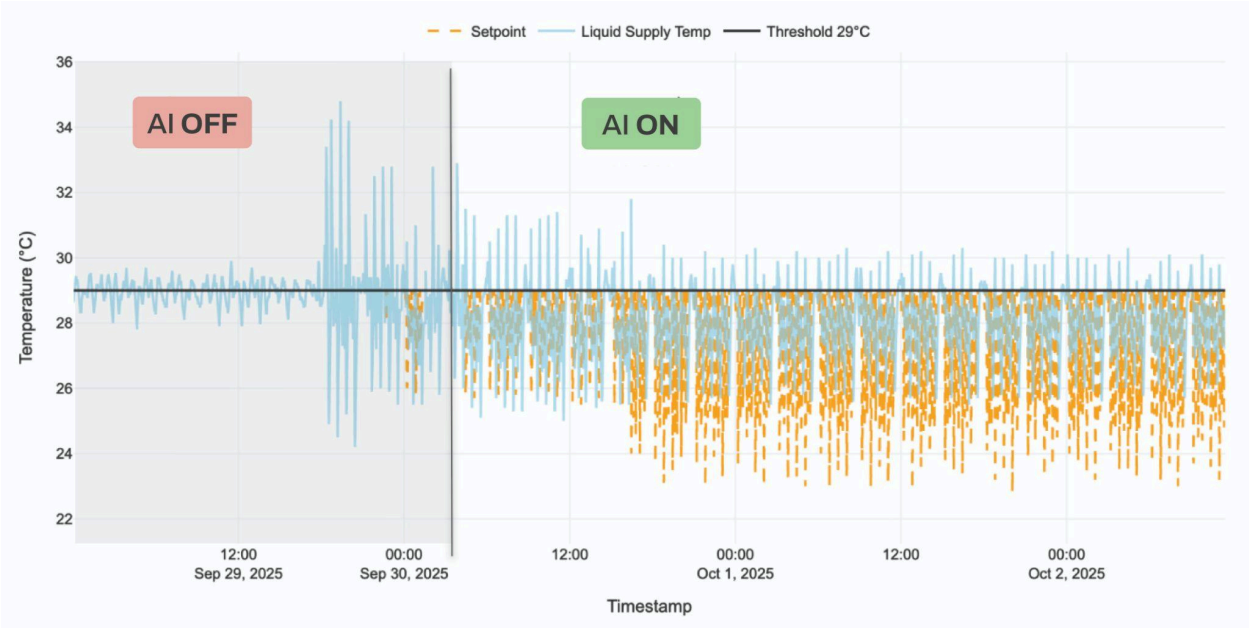

Figure 8 - A time-series graph showing AI learning on a live CoreWeave deployment (NVIDIA GB200 NVL72).

Future: Pathway to Max-Q AI Factories

The testing results (i.e. 75-80% reduction in thermal overshoot) demonstrate that proactive AI control offers a compelling approach to managing the thermal volatility arising from synchronous AI workloads. Furthermore, the AI agents were able to teach themselves to optimally manage liquid cooling CDUs without the need for human tuning and intervention.

This in turn enables AI factory owners and operators to:

1. **Maximize Tokens:** Reduce the likelihood of thermal throttling.
2. **Unlock IT Capacity:** Safely raise TCS temperatures to repurpose power from chillers to GPUs.
3. **Scale Operations:** Deploy self-tuning AI agents that continuously adapt to changing configurations and environmental conditions without manual intervention.

Looking forward, the next major unlock is operating the entire AI factory as a single unit of compute to maximize token throughput (i.e. Max-Q AI factories) across the end-to-end compute stack. This requires a deep IT/OT integration to unify the power, cooling, and workload management systems — three control planes that operate largely independently today — into a single unified data and orchestration layer for global optimization.

For Applied Digital, this convergence represents a fundamental shift in how AI factories are managed. Phaidra's AI-driven control systems enable operators to connect workload behavior directly with facility operations. Rather than operating facility systems independently from compute workloads, the company can anticipate workload-driven thermal events and dynamically optimize power and cooling delivery across the campus. This integration also enables greater operational visibility between infrastructure operators and compute tenants, allowing both parties to coordinate

around real-time system performance and maximize overall compute efficiency.

The scale and intensity of modern AI workloads necessitates this industry transition. GPU clusters have become so large and interconnected that the entire AI factory behaves as a single computer. Furthermore, in a world where revenue is constrained by power availability — and grid interconnection times are measured in years — every watt of inefficiency is a wasted watt.

In this new paradigm, we can no longer treat power, cooling, and workload management as separate entities. They must be co-designed and operated together — to maximize efficiency, reliability, and throughput across the entire compute stack. This is the purpose of NVIDIA's DSX Max-Q APIs.

Phaidra, CoreWeave, and Applied Digital are collaboratively pioneering this new frontier. Each partner brings unique strengths, capabilities, and data to bridge the IT/OT divide:

- Phaidra is pioneering AI agents for AI factories. Its agents integrate across the compute stack to provide tenants, landlords, and vertically-integrated hyperscalers with end-to-end visibility and orchestration.

- CoreWeave is The Essential Cloud for AI™. Trusted by leading AI labs, startups, and global enterprises, CoreWeave serves as a force multiplier by combining superior infrastructure performance with deep technical expertise to accelerate breakthroughs. CoreWeave has one of the largest operational footprints of accelerated infrastructure and brings deep expertise in the infrastructure space.

- Applied Digital is pioneering purpose-built infrastructure for high-performance computing (HPC) and AI workloads. It works with marquee tenants like CoreWeave to co-design AI factories for peak performance and reliability. As the landlord, Applied Digital brings deep expertise and data in OT (i.e. facility-level) operations.

Benefits of this joint operating framework, enabled by the Phaidra Prism platform, will include:

1. **Visibility for tenants into landlord-side operations:** Tenants like CoreWeave have visibility into real-time power and cooling availability at the facility-level, to plan workloads appropriately. Tenants can also orchestrate their CDU operations with the chiller plant to safely increase TCS temperatures (see Table 1).

2. **Visibility for landlords into tenant-side operations:** Landlords like Applied Digital have visibility into what is happening at the rack, server, and workload levels. This enables applications like intelligent power generation staging (i.e. pre-emptively staging onsite power generation in anticipation of large AI workloads).

3. **Agentic power allocation:** AI factory cooling systems are designed for worst-case cooling scenarios under maximum IT load. However, AI factories rarely operate under worst-case conditions, which often leaves substantial "stranded" cooling power — energy that is reserved but not actively utilized for computation. The goal of NVIDIA's DSX Reference Architecture is to

create a unified IT/OT system that provides the framework to dynamically shift power from cooling to IT (and vice versa). This deep integration will further increase overall AI factory throughput as the industry moves toward fully optimized Max-Q operations.

Acknowledgements

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